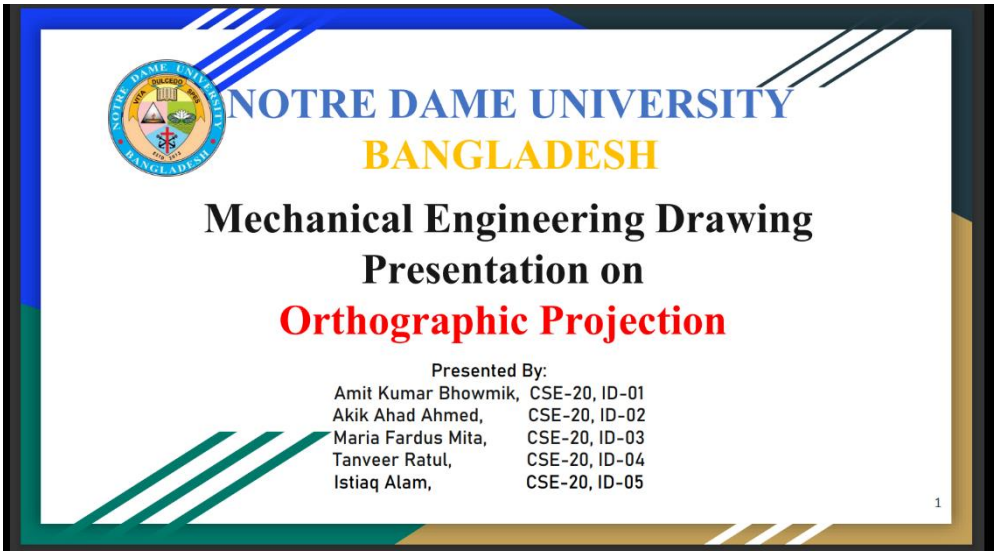
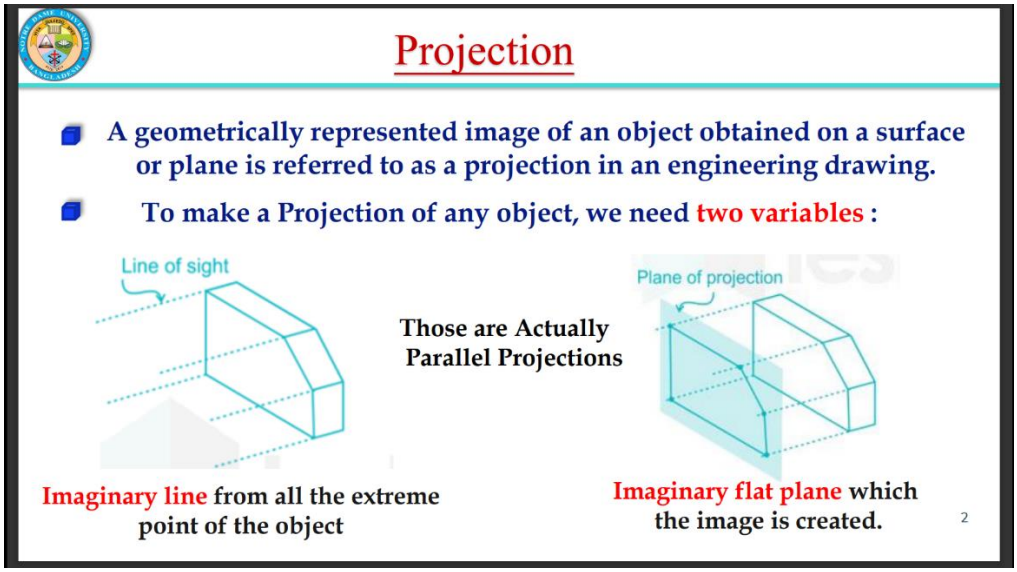
Akik:

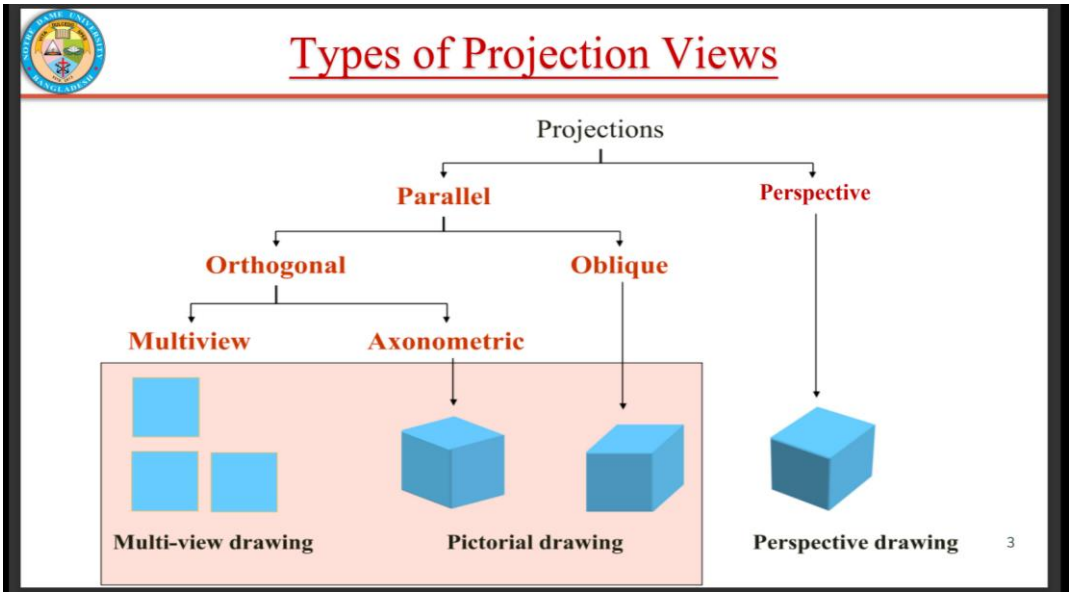


Good afternoon, everyone. This is Akik Ahad from CSE-20. I am honored to be here today to present our presentation about Orthographic Projection in Mechanical Engineering Drawing. Before we begin, I would like to Thank Dhiman Sir for giving us the opportunity. So, let's jump on the Topic!!



For Entering this Orthographic Projection, we must know about projection. So, what is projection...?? In Engineering Drawing a projection is a geometrically represented image of an object obtained on a surface or plane. There are different types of projections used in Engineering drawing, Such as parallel projection & perspective projection. But we are representing this Parallel projection here so that we can approach the orthographic projection, that is Constructed on this projection.

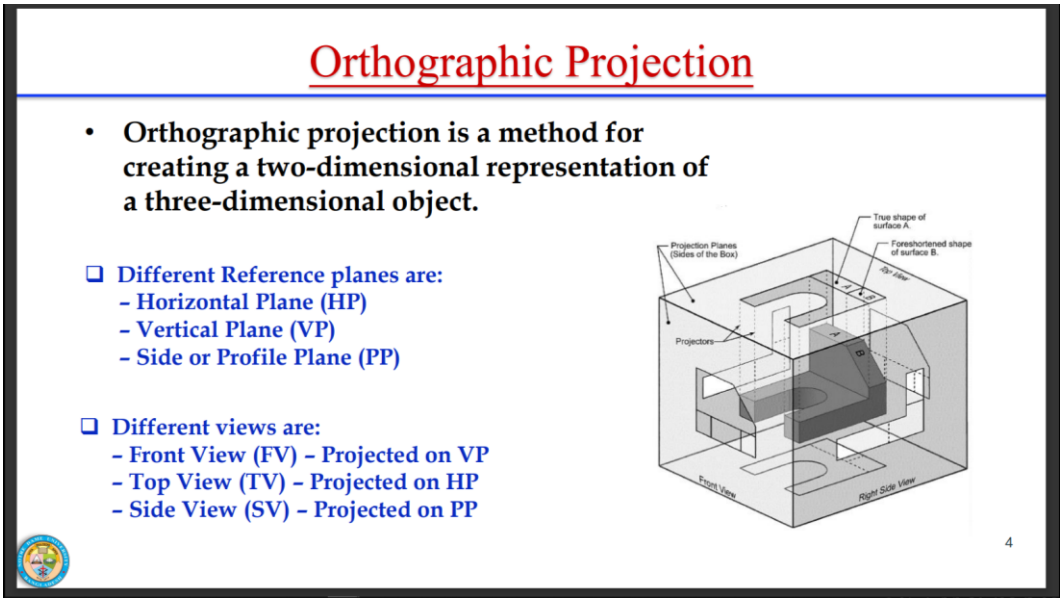
As you can see those are parallel projections, that's need two variables. Line of sight and plane of projection. In Line of sight an imaginary line from all the extreme point of the object. Then this imaginary line creates a projection that on this imaginary plan as you can see.



Here this all the types of Projection views. There are parallel projections and perspectives projection in here. According to the parallel projection we can see there is orthogonal and oblique. Under the orthogonal projection there are Multiview and pictorial drawings as their types.

Orthogonal projection is included in this parallel projection. So, my next Presenter will explain to you more about this orthogonal projection. Thank you.

Amit:

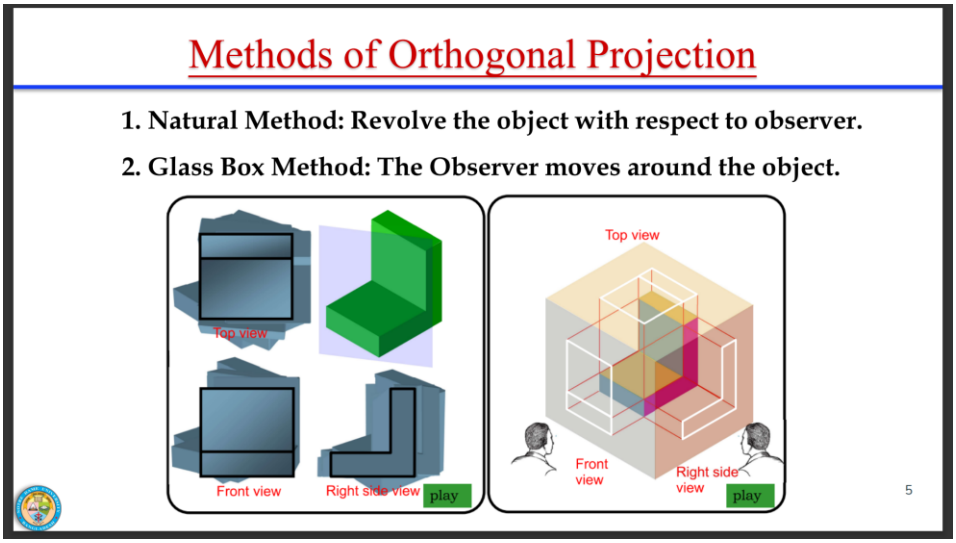


Thank you Akik. Let's take a look at this orthographic projection. Did you know what orthographic projection is!

The terms Orthographic is derived from the word "Ortho" means Right angle and "Graphic" means drawing. If the projection lines are parallel to each other and the image plane is also perpendicular to the projection lines, the result is known as an orthographic projection. Sometimes we need to represent a 3D object in a 2D interface or paper with actual dimensions & measurement.

For this kind of situation orthographic projection helps us to draw and explain any kind of 3Dimentional object in 2Dimentional plane, paper, or any kind of Graphical Interfaces.

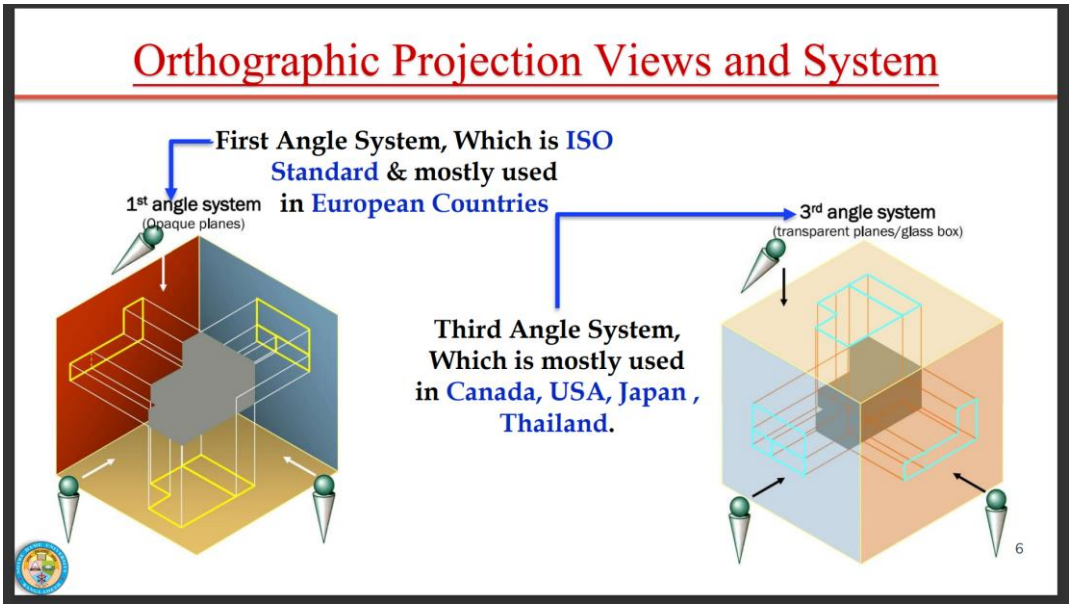
This is a projection method that reefer us different planes & different views. As you can see here, that is front view, top view & side view according to this horizontal plane, vertical plane and side or profile plane.



In the next slide here, we explain about the methods of orthogonal projections. There are two types of projections. One is the Natural method (known as first angle projection system) which is represented with respect to observer eyes. On the other hand, there is glass box method which is also known as third angle projection system.

Now my friend Maria Fardus will explain more about those methods. Thank you everyone.

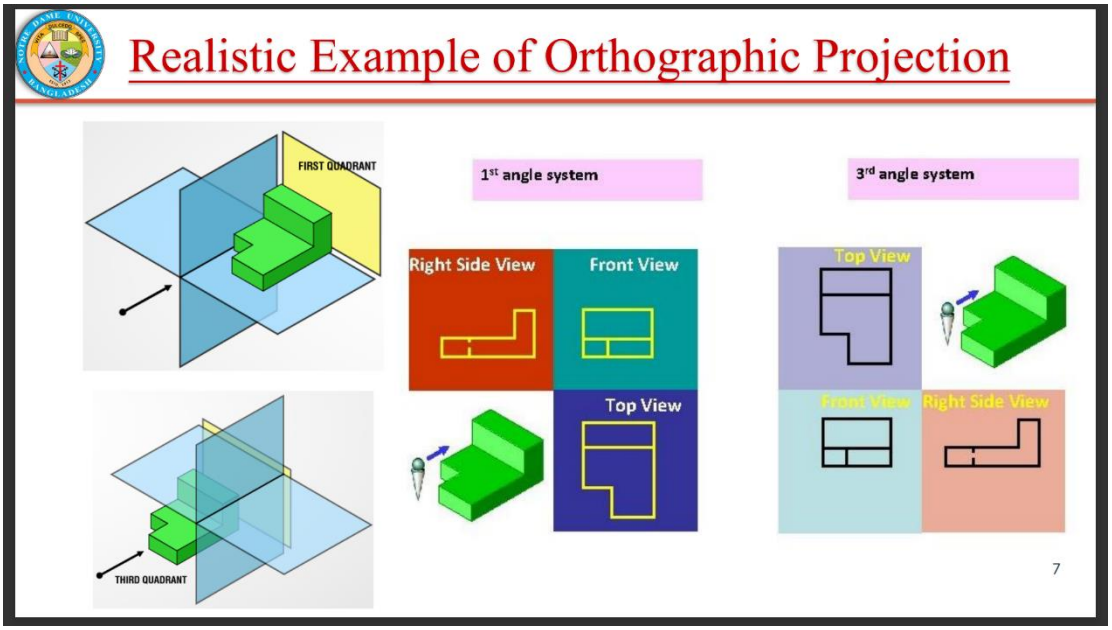
Maria:



Thank you, Amit, for giving me a chance to explain about the orthographic projection views & system. Earlier we saw details about the orthographic projection. There are two main styles of orthographic projection used:

First angle system & third angle system. First of all, in this first angle system the views of an object should be taken by placing it in the first quadrant. The characteristics of 1st angle system are front view, top view, right side view, left side view. Here the front view always comes over the top view. The top view always comes under the front view. Right side view always comes to the left at the front view. And lastly the left side view comes to the right of front view. This view is always in opposite to the observer.

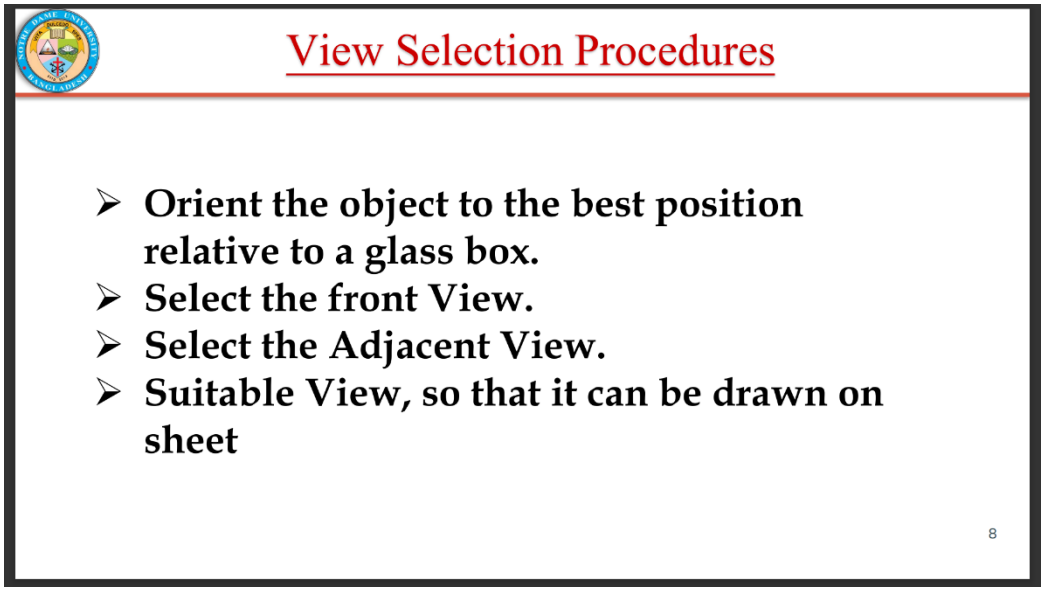
In the other hand according to the third angle system, the views of an object should be taken by placing it in the first quadrant. Here the Top view always comes over the front view. The front view always comes over the top view. Right side view always comes to the left of the front view & the left side view always comes to the right of the front view.



Here as you can see an object is shown in the slide as in isometric view.

Let’s explain this isometric drawing using orthographic projection.

Ratul:



Thank you, Maria. As We know the details about the Orthographic projection, we must know the minimum procedures for represent any isometric view of an object by this projection method.

Let's talk about the view selection procedures of this orthographic projection.

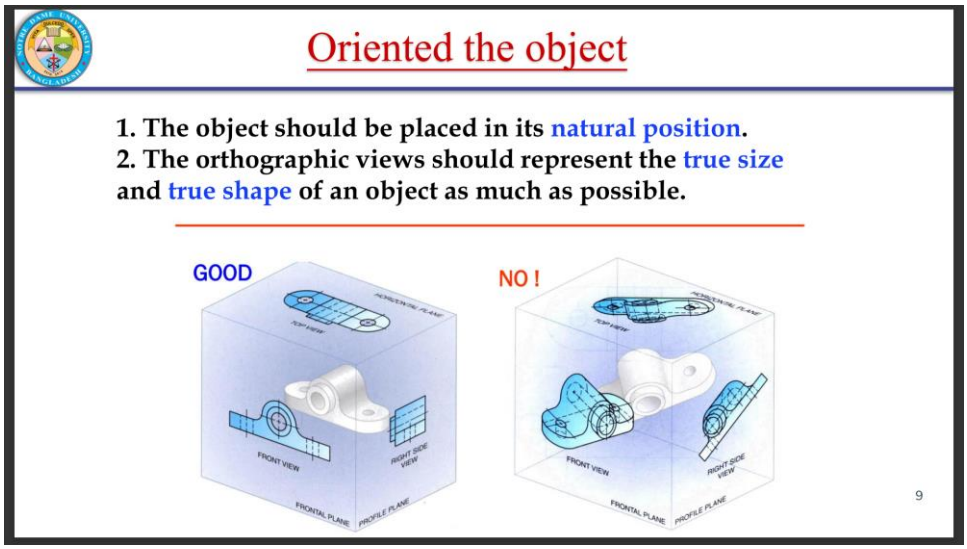
Number 1 orient the object to the best position relative to a glass box.

Number 2 select the front view.

Number 3 select the adjacent view.

Number 4 Suitable view, so that it can be drawn on paper.

As we all know, orthographic projection is used to represent any 3D object in a 2D plane or any paper or Graphical interface. For this we must orient the object perfectly as we can define which side will be the front side and which side will be the right side. This helps us to create multi view projections if we define the front side first.

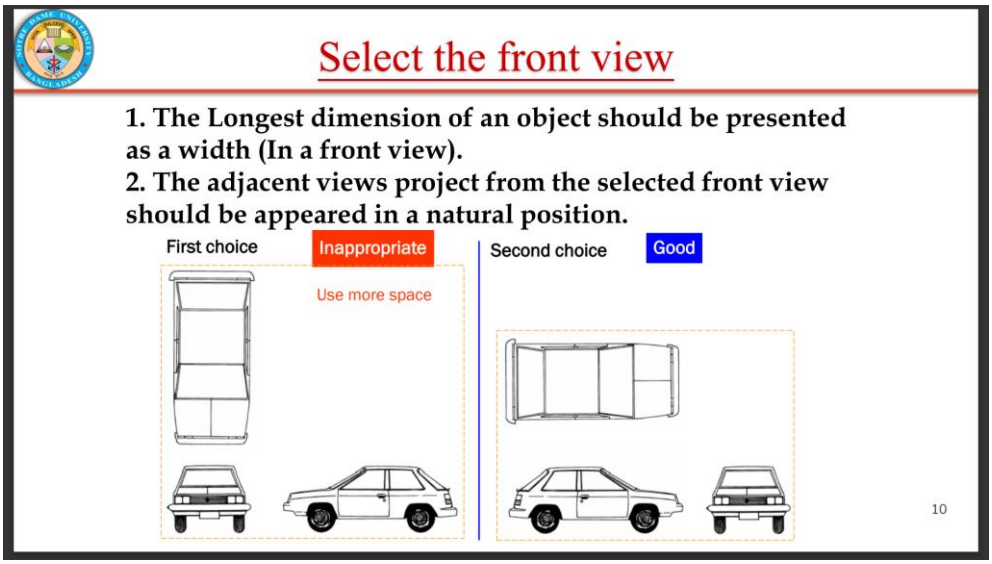


As we know this orthographic view represents the true size & true shape of an object as much as possible. So, we should place the object in its natural position. Just like this, the object into the glass boss is perfectly oriented to represent its perfect isometric view, then we can create multi views of this object.

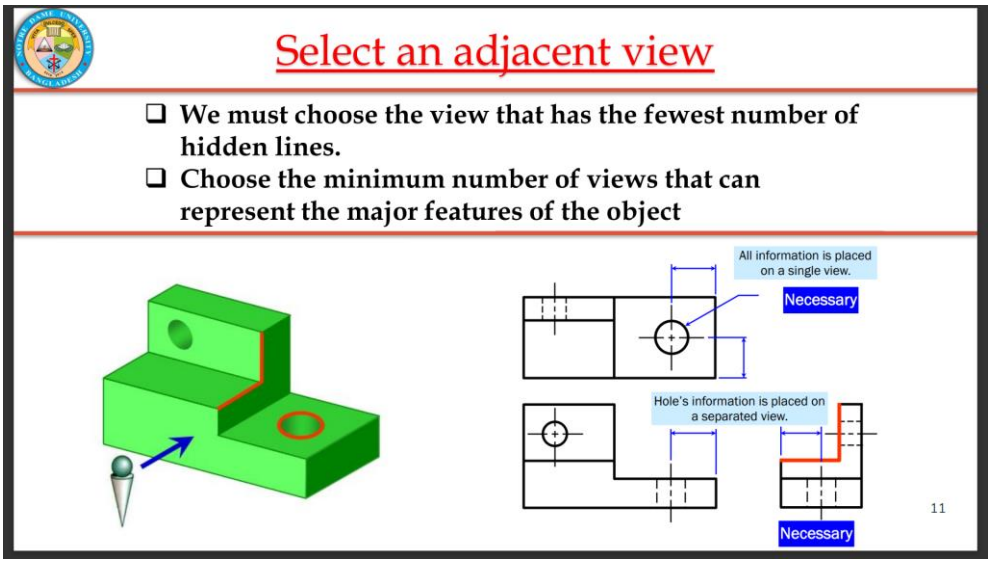
We shouldn't orient any object like this. It looks like more unnecessary & not a good practice for engineering drawing.

Now my friend Istiaq will explain to you the left-out fundamentals about this.

Istiaq:



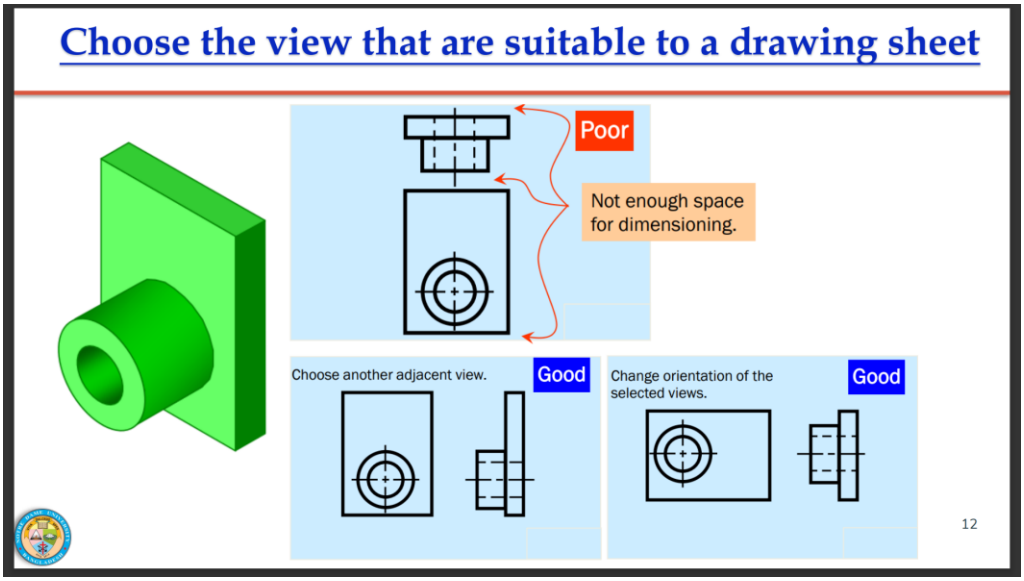
Hello everyone, thank you Ratul for giving me a chance to explain more about the view selection procedure of Orthographic projection. So, in the next fewer slides you can see the difference between the inappropriate way and good way. In the left side the object is orient like this, which is not appropriate because we cannot represent which side will be the front side of this object. In the other hand see the right-side object orientation. For the front view, the longest dimension of an object should be present as width. Secondly, the adjacent views project from the selected front view should appear in a natural position. Let's talk about adjacent views.



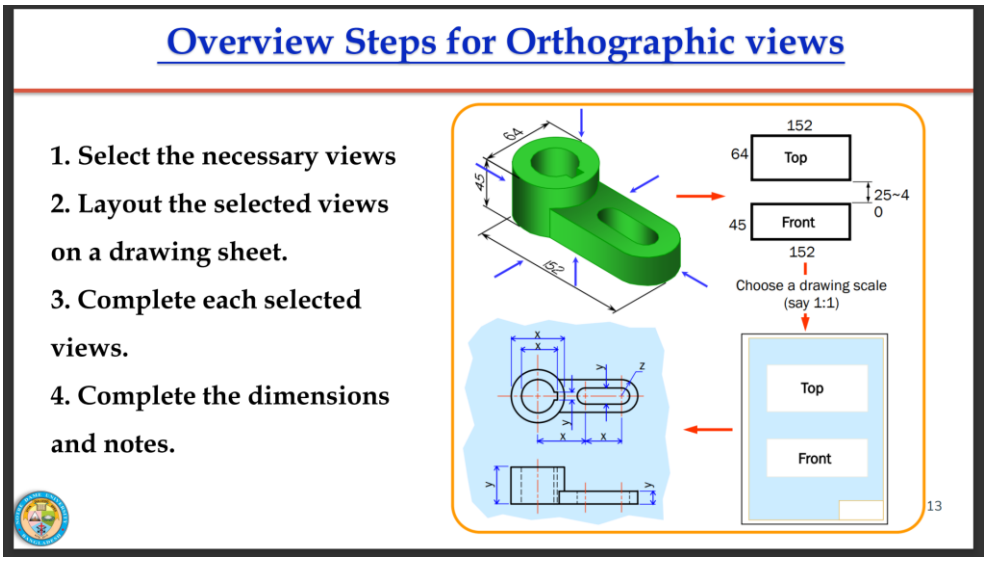
Next, we must select an adjacent view for draw the fewest number of hidden lines. As you can see on this slide, we have to choose the adjacent view wisely because it gives a minimum number of views that can represent the major features of the object.

in this view as you can see this is the top view of this object and we can explain all the information that is placed on this single view which is necessary for hole dimensions.

In the other hand hole dimension is placed on a separated view and we can explain the hidden lines here in details. So that could be the major features of the object.



Next, we must choose the view that is suitable for a drawing sheet. If we align this kind of view in our engineering drawing this is not a good practice as a professional engineer. Because here you can see in this picture there is not enough space for dimensioning the front views. Proper dimensioning & hidden lines are very important for any kind of engineering drawing. In the other figure there are proper hidden lines and space for dimensioning.



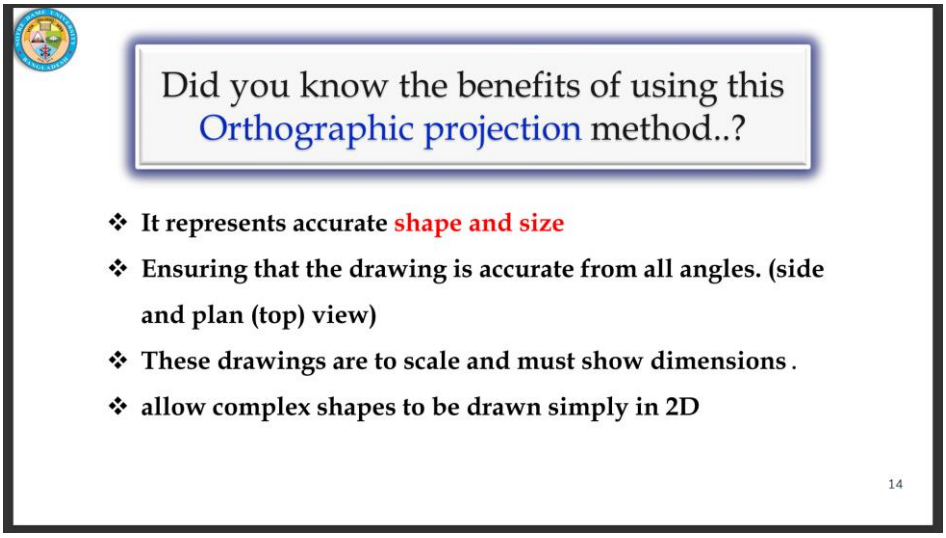
Here is the short overview steps for orthographic views.

Number 1 select the necessary views.

Number 2 layout the selected views.

Number 3 complete each selected views precisely.

Number 4 more importantly complete the dimensions and notes.



Okey. So, did you know the advantages of using this orthographic projection...?

Let me explain it for you.

Firstly, it represents the accurate shape and size of any 3D object on 2Dimensional plane, paper, or any graphical interface, which is more important in engineering drawing.

Secondly, this projection method ensures that the drawing is accurate from all angles like side view, top view, front view.

Thirdly, these drawings have proper scaling and proper dimensional system.

Finally, this projection method allows us to draw any complex 3-dimensional shapes, structures in 2-Dimensional plane or paper.

So, this is all about the orthographic projection. Thank you for your patience. Did you have any questions about this topic...? You can ask us. Now, we are starting out QnA session.

Orthographic Projection:

Suppose you want someone in another country to design this triangular Structure for you. You send them this picture, but it causes some confusion.

Do they consider the green triangle to be in the front or the back the structure? You don't speak their Language, so you can't explain it to them. What do you do?

Thankfully, we have orthographic projections to help in a situation like this. Put simply, an orthographic projection is a way of representing a three-dimensional object in two dimensions.

Orthographic projections are used to show an object from every angle to help manufacturers plan production. They can show hidden detail and all connecting parts, show all dimensions necessary for manufacture, and can be annotated to display material and finishes.

Is there anything else you would like to know?

It uses different two-dimensional views of the object instead of a single three-dimensional view. This allows you to communicate exactly what you want your structure to look like and eliminates any miscommunication between you and the person creating your design.

Typically, an orthographic projection drawing consists of three different views:

A front view, A top view and a side view.

Occasionally, more views are used for clarity.

The side view is usually the ride side, but if the left side is used, it is noted in the drawing. To draw on the views of an object, use lines to represent changes in depth.

For example, Consider this object with its right-side view orthographic projection:

Notice that there are lines where there are any depth changes in the Structure. This changes the right side of the view of a three-dimensional object into a two-dimensional picture.

What is an orthographic projection? The orthographic projection definition is explained is a method for creating a two-dimensional representation of a three-dimensional object. It is a type of parallel projection where the object is projected onto its orthogonal, or perpendicular, planes.

Orthographic projections are important because they are used to communicate the specifics of a three-dimensional object clearly and without ambiguity.

orthographic projection, common method of representing three-dimensional objects, usually by three two-dimensional drawings in each of which the object is viewed along parallel lines that are perpendicular to the plane of the drawing. For example, an orthographic projection of a house typically consists of a top view, or plan, and a front view and one side view (front and side elevations).